DID YOU KNOW?

The first Space Commission chairman Vikram Sarabhai launched the space initiative with the testing of a sounding rocket in the fishing village of Thumba, Kerala, on November 21, 1963. It was ferried on a bullock cart to the launch site

transformed ISRO into a key player in the global space community.

The SLV-3 was an all-solid four-stage vehicle weighing 17 tonnes capable of placing 40 kg class payloads in a low earth orbit. On that momentous day, the launch vehicle placed Rohini Satellite (RS-1) in orbit. It took around seven years to realise the SLV-3. Related infrastructure was required to be built in the areas of propellant, propulsion, avionics, among others, and was achieved through time-bound projects. Major strides were made in software development in the areas of trajectory studies, control system and rocket motor design and loop simulation. In all, the SLV-3 comprised 44 major functional sub-systems.

Most importantly, the SLV-3 led to the development of more sophisticated launch vehicles that could carry heavier payloads—essential for the launch of India's newer satellites. After the Augmented Satellite Launch Vehicle (ASLV) followed the SLV-3 in 1992, engineers and scientists at ISRO began work on a satellite launch vehicle capable of placing satellites into polar orbit. Thereafter, the Polar Satellite Launch Vehicle (PSLV) evolved as India's reliable and versatile launch vehicle since its first successful launch in 1994. The PSLV's performance earned it a reputation as one of the most dependable launch vehicles globally and enabled ISRO to enter the lucrative industry of launching foreign payloads from Indian soil. The Geosynchronous Satellite Launch Vehicle (GSLV) series, first tested in 2001, marks India's significant strides in launching heavier payloads into geostationary orbit.

Alongside rockets, the evolution of satellite knowhow enabled ISRO to design, fabricate and operate a slew of satellites. The development of operational satellite systems like the Indian National Satellite System (INSAT) helped India satisfy its needs in telecommunications, broadcasting, meteorology and search and rescue operations. Together, launcher and satellite manufacturing have underscored India's growing heft in the commercial space sector.

The Indian space programme of the 1980s came to be appreciated around the world as a model of cost-effective indigenous technology development. In all, these projects were distinguished by their keen sensitivity to the needs of a developing nation, and nationwide appreciation of the ability of its scientists and engineers. ■


Science

SLV-3

Soaring Into The Space Age

SATISH DHAWAN

"Walking On Peaceful Legs"

FROM THE ARCHIVES
August 15, 1980

Science **Soaring Into The Space Age**

In the course of a brief 12 minutes and six seconds one morning last fortnight, a slim pencil of steel 22.7 metres high blasted off a coastal island near Madras to launch India into the space age. "This is a great day for India and Indian science," proclaimed Mrs. Gandhi, echoing the popular view as she announced the launch in Parliament

-Amarnath K. Menon

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NO. OF LAUNCH

missions accomplished until December 5, 2024, from the Satish Dhawan Space Centre, Sriharikota, since SLV-E31 on August 10, 1979

LAUNCH SEQUENCE

- **SLV-3 launches INSAT-1A, the first multi-purpose communication satellite in 1982, INSAT 1B in 1983**
- **First successful launch of ASLV in 1992**
- **The launch of PSLV in 1997 places IRS-1D in orbit**
- **The launch of GSLV-D 1 in 2000 with the satellite GSAT-1 on board**
- **PSLV deploys 104 satellites in a single mission in 2017, a world record**



"India is the only country in which space is walking on peaceful legs. If we do not have a means to go into space how can our imaginative scientists remain alive! They will leave us"

— SATISH DHAWAN, *Chairman, Space Commission, in an exclusive interview hours after the launch of SLV-3*